



Week 6

Week 6: The End ... and the Beginning

Week 6 – the last week of the Manual programme

This week's messages:
DON'T STOP NOW!

1 Whatever stage you are at by the end of the Manual, it is doing you good.

2 Your recovery doesn't stop now – exercise, relaxation and risk factor reduction should become a permanent part of your life.

Here's what the Manual programme covers in Week 6

Congratulations on getting to the last week of the programme! This week – and every week in future – you should continue doing your regular exercise and relaxation, and maintain all the good work done so far.

Different people recover at different speeds. It isn't a matter of passing or failing by the end of the first 6 weeks – but evidence from thousands of heart attack patients over many years shows that the ones who use this type of programme can recover better than those who don't follow a cardiac rehabilitation programme.

Note carefully the two important messages in the box on the left.

We hope that you have enjoyed using the Manual and have found it helpful in your recovery. Here's to a better and healthier lifestyle than you led before.

In Week 6 of the programme:

- You will keep on building up your Exercise/Activity Plan.
- Remember to fill in your walking, exercise and activity record sheets.
- Keep listening to the relaxation CD/audio.
- We'll be looking at **hostility** and **anger**.
- We will think about going back to work. You may wish to discuss this with your GP or facilitator.
- This week's **risk factor** is **high blood pressure**.
- We shall bring together the different aspects of the **stress control programmes**.
- We shall review what has happened over the last 6 weeks and provide a checklist for future actions.

The Exercise/Activity Plan

By now you should be thinking about lifelong exercise

When you review your Exercise/Activity Plan this week, remember to think about lifelong exercise and planning for the future. If you keep on exercising, this will help to strengthen your body, reduce your risk factors and relieve stress. As the weeks and months go by, remember to keep building up the amount of exercise you are doing. At this stage the majority of people, although much improved, still do not feel 100%. If you are concerned, however, please discuss with your facilitator or GP.





Exercise/Activity Plan

Regular lifelong exercise/activity

Keeping it going

- Remember it is important to keep building up your walking or any new activity that you have taken up.
- **Do not make more than one change at a time.** This can relate to any new activity you start up. Think about changing the distance you go or the length of time you do the activity, the speed you do the activity or the amount of effort you put in.
- Remember you are aiming to breathe faster and feel a little warmer.
- Remember this needs to be an activity you can do every day.
- Keep filling in your walking record as before, and if you are doing a new activity add it in as well, and build up as your activity becomes easier.

Make it your routine

- | | |
|--|------------------------------|
| ■ Prioritise activity time | ■ Find an activity you enjoy |
| ■ Make it part of your daily routine such as walking to get the papers or go for a walk in your lunch break with friends | ■ Join a club or a class |
| | ■ Go with a friend |

Activity needs to become part of your lifestyle if you want to build and maintain the good start you have made while working through the Heart Manual. Some leisure centres have staff with specialist training and can help people to safely build up their exercise further after cardiac rehabilitation. Speak to your facilitator about referral or what is available in your area.

If you exercised regularly before your heart attack you may be thinking about how to get back to more strenuous exercise such as running, swimming, playing football or gym circuits. Speak to your facilitator for further advice or referral to a physiotherapist or exercise specialist. Remember, only do what you feel comfortable with and build up slowly.

Set Back Plan

Set backs are a normal part of life but they can interfere with following the Heart Manual Plan. These could be going on holiday, moving house, visitors or feeling unwell. What would you do? Give up all the good work you have done up till now? Hopefully not.

If it was due to holidays, moving house or visitors try and maintain as much as you can. Bear in mind you may not be doing your normal exercise/activity plan but you may be active in other ways. Once things have settled down re-write your plan and prioritise your exercise/activity. Build up gradually until you are back to where you were prior to the break.

If you are unwell stop or cut back on your walking and exercises. When you are feeling better remember to set your targets at a level which is easy and build up again. If you have been very unwell or are unsure about returning to your exercise/activity plan speak to your doctor or facilitator.



Walking record

Week 6

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

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That was	Too EasyFairly EasyFairly HardToo Hard

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard

Date	I think I can easily
That was	Too EasyFairly EasyFairly HardToo Hard



Exercise record

Week 6

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	

Date	How long exercised	How many times
That was		
	Too Easy Fairly Easy Fairly Hard Too Hard	





Daily activity record

Week 6

Write down the things you do and how you feel each day

Date
Date
Date
Date
Date
Date
Date



Do you ever get so angry you could explode?

The story so far ...

Over the past weeks we have looked at several different kinds of stress:



Time pressure



Overwork



Speeding



Driven behaviour

We have seen how stress affects us by raising our adrenaline level. Adrenaline is a chemical produced by the body.

At normal levels, or in emergencies, adrenaline is very useful (if you want to know how it helps in emergencies, look at pages 152–162 in Part 3 of this Manual). Adrenaline becomes a problem when we have high levels in our blood all the time. This makes us feel tired and sick, and affects our mood.

Stress alone isn't a major risk factor

BUT ...

At present there is some evidence to suggest work stress can have a small effect on later development of coronary artery disease.

BUT it's possible that people who tend to experience anger, annoyance or irritation more frequently or intensely than others may be more likely to build up the risk factors which lead to coronary artery disease. And it may be that these feelings can affect recovery. If you sometimes 'blow your top' or store up a lot of intense emotions inside, this section might help.

Hostility doesn't mean punching someone

Psychologists call these feelings **hostility**. Having hostility doesn't mean that you are going to become violent or hurt someone. It means that you have some strong emotional feelings.

The BAD NEWS about hostility

People with a lot of hostility can:

- mistrust and blame others when frustrated
- feel angry with those they blame
- behave aggressively.

Hostility can interfere with happy and healthy relationships. It can lead to social isolation. It can interfere with your ability to listen to and take advice from your loved ones and your healthcare providers.

People with high levels of hostility tend to have greater increases in adrenaline when they get stressed. They also have more difficulty in relaxing afterwards.

The GOOD NEWS

There are lots of things you can do to cope with hostility.

Coping effectively with hostility can:

- **improve your physical health**
- **decrease your stress**
- **make you less irritable**
- **enhance your relationships with others**
- **improve the overall quality of your life.**



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Dealing with hostility and anger

You can learn to control hostility and anger. Here are some tips.

1 Learn to identify your triggers and reactions

Keep a diary of situations in which you feel angry, frustrated or annoyed.

2 Challenge your automatic thoughts

Identify your 'negative automatic thoughts' – such as when you immediately assume the worst or find fault or blame others. Then challenge these automatic thoughts by examining the facts that support them and the facts that don't support them.

This is what Meena Vyas did. While driving to work, she was cut in on by another driver and became instantly enraged. 'I could kill that selfish, arrogant **', she thought (fill in the blanks yourself). Then she took a step back from that thought and challenged it by looking at the facts that might contradict her automatic thought:**

'How do I know that the other driver was being selfish or arrogant?' 'Maybe he made an innocent mistake and didn't see me.' 'Maybe he's in a rush or late for something important – like I am sometimes' or 'Maybe he's a new driver without much experience.'

3 Decide how important the situation is

- Do you have all the facts?
- Is it worth your time and effort to make a big deal of the situation?
- Can you do something to change or resolve the situation?

Meena decided that there was nothing she could do. The situation was not critical. She reminded herself how lucky she was to have survived her heart attack. In the grand scheme of things, being cut in on by another driver really didn't matter all that much.

4 Use behavioural strategies

When you feel unable to control the intensity of your frustration, annoyance or anger, take a 'time out.' Take a break. Remove yourself from the situation. Take a big breath, relax, clear your mind, and get the situation back into perspective.

Practise saying No. Turn back to pages 92–93 and re-read how Wendy Taylor reduced her stress levels by saying No and making her own decisions about what she would and wouldn't do.

5 Use relaxation techniques

Keep using the relaxation techniques you have been practising over the last 6 weeks. Stopping smoking and cutting down on caffeine, alcohol and sugar intake can also help. So can exercise.

6 Be positive

Distract yourself from upsetting thoughts, situations and feelings. Try to find a little humour in every situation.





Week 6

This week's risk factor: High blood pressure

Why does high blood pressure matter?

Blood pressure is like cholesterol – we all have it, but some people have more of it than others. Doctors have found that people whose blood pressure is at the top end of the range have an above-average risk of getting coronary artery disease and other circulation problems. So blood pressure matters because it's a **risk factor**.



How do I know if I've got high blood pressure?

From measuring blood pressure in thousands of people for many years, doctors have been able to work out levels for 'normal' or 'high' blood pressure. Because your heart pumps in beats, two blood pressure measurements are taken – a peak figure when the heart beats, and a lower one when it relaxes. Your doctor or nurse will express this as two numbers such as 130 over 80. It is important that you have your blood pressure checked regularly so that you know what your blood pressure should be. Remember that these figures are for when you are resting, and it is normal for blood pressure to go up and down depending on your activity.

What causes high blood pressure?

For most people with high blood pressure there is probably no special cause, but there are things which tend to put the blood pressure up into the 'high' range.

These include **being overweight, lack of exercise, drinking too much alcohol/smoking and kidney disease. Too much salt** in your diet can also raise your blood pressure. Some doctors think that stress can raise it, too.

What can be done about it?

There's good news! There are many medicines that can lower blood pressure, and the risk of developing coronary artery disease decreases at the same time. And if you or your doctor thinks that your blood pressure is caused by any of the factors listed above, you can do your bit to help by:

- reducing your alcohol intake
- losing weight by dieting
- taking less salt in your food
- practising your relaxation – even if you don't feel stressed
- taking regular exercise
- stop smoking
- take your medication as prescribed
- attend GP surgery appointments for regular checks of your BP.



Back to work

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If your heart attack has kept you off work, you may be wondering when it's safe to go back. If you have a non-manual occupation, such as office work, you can think about returning to work from about 4 weeks after your heart attack. You may even be able to do some work from home prior to that, but remember to take things slowly at first. Many people find that their concentration, memory and sleep can be affected in the early stages of their recovery. If you have a heavy manual job you may not be able to return to work until later. Returning to work can be an anxious time, but many people who return after a heart attack are doing their full range of duties within a few weeks.

Here are some tips which might help.

- Get advice first** When you feel ready to go back to work, talk to your doctor or facilitator first. They can advise you whether it's safe to go back yet.
- Contact your employer** Get in touch with your employer and discuss your plans for coming back. Most employers will be very understanding and will let you do shorter hours with lighter duties at first.
- Build up your working hours gradually** It's probably not sensible to go straight back to exactly the same work routine as you had before your heart attack. Plan to start with fewer working hours and gradually build up to your normal hours over an agreed number of weeks. For example, you could:
- take work home initially – this might be a good option if you are a desk worker, manager, etc.
 - start with either mornings or afternoons, or
 - spend alternate days working and not working
 - then increase your hours gradually until you are back to normal.
- Start with lighter duties** Make a list of all the tasks that you do in your job – daily, weekly, monthly and one-off jobs. Put them in order from the easiest to the most difficult.
- Discuss and agree with your bosses that at first you will only do the lighter duties. Over an agreed timescale you can gradually get back to doing the harder ones.
- Make sure that you get feedback** Ask your employer to name someone (like a manager or foreman) who will be able to give you regular feedback on how you are doing, and who can decide with you which extra tasks you should take on and when.
- Take all your rest periods and breaks** Try not to miss rest periods and meal breaks. If the job will allow, it's a good idea to take one or two extra rests during the course of the day.

<u>Daily Tasks</u>	
Order processing	1
Answering phone calls	1
Filing	2
Post Office	4
Packing and despatch	5
<u>Weekly Tasks</u>	
Stock check/ordering	3
Local van deliveries	4
Unpacking stores	5
<u>One-off Tasks</u>	
Sort cabinets	3
Control systems	4
Re-organise store room	5

When you go back to work, try to remember some of the things you have read about in previous weeks of this programme. If you think your old ways of working may have been adding to your risk factors, then you will probably agree that it wouldn't be a good idea to let yourself go back to them. Don't forget that the Manual is here to help.

- Re-read the sections about **time pressure, deadlining**, the **Superhero complex**, the **Yes complex** and **stress at work** on pages 88–95.
- Re-read the sections about **speeding, overworking** and **driven behaviour** on pages 106–113.
- Try to get regular exercise during the working day.
- Remember to do your relaxation while you are at work.



Week 6

What has happened over the last 6 weeks

Over the last 6 weeks we have covered all of the most important things that you need to know about and to do after a heart attack.

Information:

- what a heart attack is and what causes it
- what coronary artery disease is and what causes it
- wrong ideas about heart attacks and coronary artery disease
- how to fight back and reduce the risk of having another heart attack
- what to do if you think that you may be having another heart attack.

Exercise

You have been helping your recovery and improving the strength of your body by using the Exercise Plan.

Relaxation

By now you should know from the Manual and the relaxation CD/audio about all the different ways in which relaxation can help you.

You should be able to notice tension developing and have ways to reduce it. Make these a part of your life without the relaxation CD/audio.

Stress reduction

You have looked at some of the ways in which stress can affect you, and what you can do about it.

Risk factors

You have looked at the main controllable risk factors and how to control them. These risk factors are **smoking, poor diet, being overweight, lack of exercise, high blood pressure and stress.**

Other worries

You know that having a heart attack doesn't mean that you have to give up the more enjoyable things in life, like alcohol, eating out and sex.

You may also have used Part 3 of this Manual to find out more about:

- medicines – what they do for you and their side-effects
- treatments you may have had or may be waiting for
- anxiety after a heart attack – why it happens and how to cope
- stress in relationships
- angina, other chest pains and palpitations
- low spirits and how to reduce them.

Coping with setbacks

By now you may be feeling fully fit. You may be wondering what all the fuss was about. That's fine, so long as you don't forget about the things you need to do to protect yourself from having another heart attack.

Or you may still be feeling less than 100%. That's only to be expected, so long as you don't let it get you down.

There is no 'normal' time for recovering from a heart attack. Indeed we wouldn't expect some people to be fully fit for some months yet.

Week by week you will be getting stronger.

Sometimes you may feel as if you have stopped making progress. It's only when you look back a few months later that you can see that this isn't the case.

If it begins to worry you, look back over your activity or exercise sheets or discuss your feelings with your facilitator. See how much more you can do now compared with what you could do before.





Week 6

What has happened over the last 6 weeks (continued)

Coping with setbacks (continued)

You may have been left with some breathlessness or angina. You may even have had to go back into hospital.

If so, there is no reason for despair. Everything in the Manual remains just as true.

Read through the true facts about heart attacks and coronary artery disease again to remind yourself.

If your recovery has not been as quick as you had hoped, this doesn't mean that the outlook is bleak. Far from it.

The fact is that now the doctors know about your condition, you are much safer than you were before your heart attack.

There are many other things that can be done to cope with the complications that sometimes arise. New medicines and other treatments have revolutionised the treatment of coronary artery disease.

Finally, you may like to try this checklist

How many of these things are you doing to protect yourself?

I am reducing my risk factors by:

- ☐ stopping smoking
- ☐ reducing smoking
- ☐ reducing fat consumption
- ☐ losing weight
- ☐ eating less
- ☐ exercising more
- ☐ being more active (less sedentary)
- ☐ managing my stress levels

I am lowering my blood pressure by:

- ☐ relaxing
- ☐ losing weight
- ☐ reducing salt intake
- ☐ exercising more

I am eating more:

- ☐ fruit and vegetables
- ☐ fish
- ☐ wholemeal bread
- ☐ high-fibre cereal
- ☐ beans, pulses and lentils
- ☐ polyunsaturated fats
- ☐ rice, pasta and potatoes

I am eating less:

- ☐ red meat
- ☐ bacon
- ☐ crisps, cakes and biscuits
- ☐ fried take-aways
- ☐ butter
- ☐ whole-fat cheese
- ☐ sweets and chocolate
- ☐ sugary drinks
- ☐ and I'm drinking less alcohol

I am doing regular exercise:

- ☐ walking more
- ☐ cycling
- ☐ exercise classes
- ☐ other interesting and enjoyable exercise
- ☐ and I'm planning regular exercise or sport sessions in future

I relax regularly by:

- ☐ using the methods on the relaxation CD/audio
- ☐ using my own method
- ☐ using imagery
- ☐ using breathing exercises
- ☐ making regular times to relax
- ☐ finding new hobbies or interests

(continued overleaf)



Week 6

What has happened over the last 6 weeks (continued)

I use rapid relaxation:

- ☐ when driving
- ☐ when angry, worried or fed up
- ☐ to relieve angina

I work on reducing time pressure by:

- ☐ spending more time with family and friends
- ☐ making time for enjoyment
- ☐ using deadlining
- ☐ making regular time for myself
- ☐ saying No
- ☐ dropping the Superhero complex
- ☐ charging more for my time

I am reducing speeding by:

- ☐ getting control of my workload
- ☐ using relaxation
- ☐ using deadlining
- ☐ not putting things off
- ☐ taking things more slowly and steadily
- ☐ doing slowing-down exercises
- ☐ remaining calm when waiting

I am reducing overworking by:

- ☐ getting control of my workload
- ☐ increasing my other interests

- ☐ using relaxation
- ☐ changing my attitude to enjoying myself
- ☐ sorting out home tensions
- ☐ reducing my needs
- ☐ getting out of the habit

I am reducing driven behaviour by:

- ☐ changing my beliefs about success
- ☐ listing my achievements and ambitions

Other things:

- ☐ I am taking my medication as prescribed

It's quite a list, and we don't imagine that you are going to manage to do all of these things, or that you **need** to do all of them.

But it does show just how many things **you can do** to fight back against coronary artery disease.

If you have other medical conditions, think about how the changes you have made have improved these aspects of your health. Has it made it easier to move about if you have arthritis, lowered your blood sugar, or helped you sleep better?

The changes you have made will help you stay healthier and improve many aspects of your general well being, not just your coronary artery disease. Continue to build on the improvements you have made for a happier and healthier future.

The Future

We hope that you have enjoyed using this Manual, that it has answered some of your questions and that it has helped your recovery. We also hope that the book and the CDs/audio will go on being useful in the following months. Please note: the internet can be a great source of information, however please ask your facilitator or GP which web-sites they recommend.

Please fill in your End of Programme Questionnaire on pages 167-170 and send it (along with your Baseline Questionnaire) to the Heart Manual Department.



Part 3

Facts and Advice to Help Your Recovery

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Medicines

Like most people who have had a heart attack, you have probably been given several different kinds of pills to take home with you.

The doctor or nurse may have explained to you what they all were and why you have to take them.

If you are like most patients in the same situation, you were probably too concerned about going home to take it all in at the time.

So we have included this section in the Manual to answer some of the most common questions you might have about your medicines.

I feel much better – so why do I need to take medicines?

Doctors prescribe medicines after a heart attack for two reasons:

- 1 to reduce your chances of having another heart attack in the future **or**
- 2 to treat any problems caused by the heart attack you have just had.

We shall deal with each of these in turn.

1 Reducing your risks for the future

In Parts 1 and 2 of the Manual we explained how a heart attack happens and we discussed the risk factors that lead to coronary artery disease.

We explained what you can do to reduce your risk of future problems. But there are also many things that your doctor can do to help.

For instance:

- In Part 1 we explained that a heart attack is caused by a clot forming in an artery which has coronary artery disease. One way to make this less likely to happen in future is to make your blood less 'sticky', so that clots don't form so easily. The simplest and most effective way to do this is to take an aspirin every day.
- Some tablets have been shown to reduce the risk of future heart attacks, although doctors are unsure how they do this. For instance, we know that if we give beta-blockers to patients who have had a heart attack, these medicines can reduce the chances that they will have another one.

Some tablets may be prescribed to help to control risk factors such as high cholesterol or high blood pressure.

2 Treating problems caused by your heart attack

By 'problems' we mean:

- **symptoms** which you complain of, such as chest pains or breathlessness (more about this later)
 - **signs** which doctors find when they examine you, such as fluid in your lungs or ankles, or signs which they find on your tests, such as an enlarged heart or irregular heart rhythm.
- We have medicines to treat many of these things.



Medicines (continued)

So all these tablets don't mean that I'm really ill?

Not at all. Nowadays it is quite normal to be taking at least four medicines after your heart attack. **The number of pills you are given has very little to do with the size of your heart attack or the amount of disease in your arteries.**

What about side-effects?

All tablets have side-effects, but most of these are not serious. For instance, almost all medicines are capable of causing rashes, which should be discussed urgently with your doctor. Any other side-effects, such as stomach upsets, headaches, etc., should also be discussed with your doctor or facilitator.

There are good reasons why you have been given each of the pills that you are taking, but if a pill makes you feel ill, your doctor may prescribe something else. There's always a choice, so don't be afraid to ask.

Can I take supplements?

Some over the counter medicines, herbal remedies, supplements and even foods like grapefruit or leafy greens can affect how certain medicines work. In some cases they can make the drug work faster, make it stronger, or alternatively not work as well as it should.

It is important to tell your doctor or pharmacist if you are taking, or wish to take any supplements. Read the information leaflet that comes with your medicine and take it as your doctor prescribes.

If I don't like one of the pills, can I stop taking it myself?

Absolutely not!

It can be dangerous to stop some pills suddenly, because the body needs time to adjust to doing without them. If you really want to stop taking them, discuss this with your doctor or facilitator. It may be possible for you to take fewer pills or lower doses, and together you can agree a plan for this.

What if I forget to take them?

Missing the odd dose probably won't cause serious problems.

If it is a pill that you take once a day, there will still be plenty of the drug in your body the next day, so just take your next dose as usual.

If it is a pill that you take two or three times a day, the next dose will be due soon and will 'top up' your levels then. If you are concerned, contact your doctor or facilitator.

The best thing is not to forget your pills in the first place. Make a list of which medicines to take at which times, and connect it with something which you usually do at that time – such as your morning wash, eating lunch, getting changed after work, or going to bed. If you always take your pills at these times it will soon become a habit.

I find all the names confusing

Drug names **are** confusing.

The information on the next few pages may be helpful.



Medicines (continued)

How drug names work

Any pill might be described in three different ways.

Here is an example.

a Beta-blocker

The kind of drug it is.

b Atenolol

The **name of the chemical** in the drug.

c Tenormin

The **trade name** (usually begins with a capital letter).

Different companies sell the same chemical under different trade names, so one chemical may have more than one trade name.

Below are some of the most common kinds of drugs used. It isn't possible to list every drug, so if one of your pills isn't mentioned, ask your doctor what kind it is.



1 Antiplatelet agents

These medicines are the real risk reducers- probably the most important pills you can take after your heart attack.

Aspirin is the drug most commonly used to make the platelets in the blood less 'sticky'. This reduces the likelihood of clots forming in, and then blocking diseased arteries.

A low dose of aspirin a day can markedly reduce the risk of another heart attack or stroke.

We know that this risk is reduced even further if aspirin is taken alongside another type of antiplatelet agent. Examples of these additional antiplatelet agents are:

- **ticagrelor**
- **prasugrel**
- **clopidogrel.**

Taking aspirin **and** one of the drugs above is called dual antiplatelet therapy (DAPT). You may be given this treatment for a period after your heart attack. Your doctor will decide if this is the right treatment for you and how long you should take both tablets. This will depend on any other medical conditions you may have and if you have any issues whilst taking the tablets. If you have no problems taking the low dose aspirin, this will be a lifelong treatment.

If you have asthma or problems with your stomach, taking aspirin may not be possible, or you will need to take an additional pill to protect your stomach.

A side-effect of making your blood less 'sticky' is that you may bleed more easily, but the benefits of taking these tablets

Medicines (continued)

are important in managing your risk after a heart attack. Only take the doses prescribed, more is not better!

2 Beta-blockers

These medicines are another risk reducer, particularly during the first year or so after your heart attack. They may also be used to control symptoms such as angina.

They block the effects of adrenaline, slow down the heart rate and make the heart muscle work less hard.

The chemical names of beta-blockers usually end in '-olol.'

Examples of beta-blockers include:

- **atenolol**
- **bisoprolol**
- **metoprolol.**

When you first start on a beta-blocker your dose may be slowly increased by your doctor. This is normal. You should tell your facilitator or doctor if you have any side-effects.

Side-effects can include tiredness, cold fingers and toes, wheeziness (these drugs should normally be avoided in patients with asthma), nightmares, and difficulty with erections.

These side effects can often be dealt with by reducing the dose or changing the particular type of beta-blocker after consulting with your doctor.

You should not stop beta-blockers suddenly, as this may cause severe angina. Make sure that you always have a good supply.

3 Statins

These medicines are yet another risk reducer!

You know that high cholesterol is a risk factor for coronary artery disease. For many patients, lowering their cholesterol level will reduce the likelihood that they will have another heart attack.

Statins are given routinely to all patients who are known to have coronary artery disease, even if your cholesterol level is within the normal range. Statins work by reducing the amount of cholesterol made in your liver, but there is increasing evidence that they also have some anti-inflammatory effects which benefit the heart.

Examples of statins include:

- **atorvastatin**
- **rosuvastatin**
- **simvastatin.**

These medicines work very quickly and lower your risk within weeks especially when combined with changes to your diet.

Side-effects can include tiredness, muscle pains, stomach upsets and headache.

Blood tests will be done to monitor the effect of the drug on your cholesterol level.

Occasionally these medicines can cause inflammation in the muscles or liver. This may be found on blood tests. You should report any new muscle pain, tenderness or weakness immediately to your doctor. If this happens your doctor may

change you on to a different statin, ask you to take a lower dose, a dose every other day, or to stop the tablet for a time and then restart it. These steps often help overcome any initial side effects.

You may be advised to avoid eating grapefruit or drinking grapefruit juice as it can interact with some statins increasing the chances of some side-effects.

Sometimes making diet changes and taking a statin is still not enough to manage your cholesterol levels. Your doctor may prescribe other medicines to take alongside your statin or refer you to a specialist clinic.

These work in a different way.

4 Angiotensin-converting enzyme (ACE) inhibitors and angiotensin-receptor blockers (ARB's)

These can reduce your risk, too!

These medicines relax the blood vessels and can lower your blood pressure.

After a heart attack, they can improve the pumping action of the heart.

This may help breathlessness or tiredness, or simply help the healing process.

The chemical names of ACE inhibitors usually end in '-pril.'

Examples of ACE inhibitors include:

- **lisinopril**
- **ramipril.**

When you first start on an ACE inhibitor, your dose will be



Medicines (continued)

slowly increased by your doctor to allow your body to get used to the effects of the drug, this is normal practice. Your doctor will check your blood pressure and take blood tests. Remember to tell your doctor if you have any side-effects, such as, dizziness when standing, persistent cough or taste disturbance.

ACE inhibitors work better for some people than others. If it doesn't work for you, or you are unable to take an ACE inhibitor, you may be prescribed an angiotensin-receptor blocker (ARB) instead. Like ACE inhibitors, ARB's relax the blood vessels and lower blood pressure.

The chemical names of angiotensin-receptor blockers usually end in '-sartan'.

Examples of ARB's include:

- **losartan**
- **candesartan.**

5 Nitrates

These medicines are used to control angina.

Nitrates work mainly by relaxing the veins that take blood into the heart. This has the effect of reducing the amount of blood returning to the heart, so that the heart has less work to do. These medicines may also relax the arteries, particularly the coronary arteries, improving the blood supply to your heart muscle.

On leaving the hospital you may have been given a short acting **glyceryl trinitrate (GTN) spray**, which you squirt under your tongue if you have chest pain. If so, you should also have been given advice on how to use it.

The usual advice is to use the spray and wait for 5 minutes. If you still have the pain, repeat the dose 5 minutes later.

If however your pain is unbearable or getting worse, or if you still have pain 5 minutes after the second dose, dial 999 or your local emergency number and ask for an ambulance. Tell them it is a suspected heart attack. If in doubt, read page 35 in Part 2 of the Manual.

If you have angina, you may be advised to take your GTN to prevent episodes occurring.

Read page 145 for advice on controlling your angina.

There are also other forms of GTN such as short or long acting tablets.

Examples of these are:

- **glycerol trinitrate**
- **isosorbide mononitrate.**

Side-effects can include dizziness when standing up and headaches. The headaches tend to disappear after the first couple of weeks of use.

Some GTN tablets only last 8 weeks once the pack is opened so if you use GTN tablets keep track of when you opened them and make sure you get a new supply after 8 weeks.

6 Calcium-channel blockers

You may be given a calcium-channel blocker if you are not able to take a beta-blocker.

These medicines help treat angina and high blood pressure, and occasionally are given for palpitations or heart rhythm irregularities.

Like nitrates, they relax the blood vessels, but they can also act more effectively on the arteries leading from the heart. They make it easier to pump blood out to the body, reducing the blood pressure. They can also relax the coronary arteries to increase blood flow to the heart muscle. In some cases they relax the heart muscle itself and slow the heart rate.

Examples of calcium-channel blockers include:

- **amlodipine**
- **diltiazem**
- **verapamil.**

Side-effects can include flushing of the face, ankle swelling, nausea, constipation and headache.

These medicines are used to treat angina.

They relax the arteries and veins, thus reducing the work that the heart has to do. They also relax the small arteries in the heart, thereby helping to improve the blood supply to the heart muscle.

An example of a potassium-channel activator is:

- **nicorandil.**

Side-effects can include headache, dizziness on standing up, nausea and flushing of the face.

Ivabradine can be used if beta-blockers and calcium-channel blockers are contraindicated (not suitable) as they work by slowing the heart rate down.

7 Diuretics

These medicines are used to remove excess water which can build up in the body after a heart



Medicines (continued)

attack and cause ankle swelling or breathlessness.

Some of them can be used to treat high blood pressure.

They work on your kidneys to make you pass more urine.

Examples of diuretics include:

- furosemide
- bumetanide
- indapamide
- eplerenone.

Side-effects can include losing too much fluid, leading to feelings of tiredness or dizziness when you stand up. Other effects can include muscle cramps, headache or feeling sick.

It is important to try to reduce your intake of salty foods such as bacon, cheese, crisps, peanuts and Bovril or Marmite, which can encourage the build-up of fluid.

As diuretics make your kidneys work harder your doctor will want to monitor them by occasionally taking some blood tests.

8 Other kinds of drug

Some patients require medicines to treat irregularities in the heart rhythm. These are known as **anti-arrhythmics**.

Your cardiologist will advise you how long you may need to take these.

Examples of anti-arrhythmics include:

- digoxin
- amiodarone
- verapamil.

These medicines are all very different to each other, so be sure to read the information leaflets to check for side-effects.

Anticoagulants work to 'thin the blood' meaning it stops the blood from clotting so easily. Warfarin is one of these drugs and has been used for many years. There are also newer types of anticoagulants called Direct Acting Oral Anticoagulants (DOAC's). These include:

- rivaroxaban
- apixaban
- dabigatran
- edoxaban.

These can be given after your heart attack if you have suffered complications which caused clots to form in your veins or inside your heart, or if you had problems with your heart rhythm soon after the heart attack.

If you have continuing problems with your heart rhythm or with

the valves inside your heart, you may need anticoagulant treatment for life.

As anticoagulants 'thin the blood' you may bruise easily and bleed more from cuts. If this gets worse over time you should inform your doctor. **The side-effects from anticoagulants can be serious but there are important benefits to taking them** and you should speak to your doctor if you are worried.

You should avoid aspirin if you are taking an anticoagulant, unless advised by your doctor. You should also check with your pharmacist or doctor before taking over the counter medicines, herbal remedies, or supplements as some can interfere with how the anticoagulant works.

If you are taking warfarin, you will need regular blood tests to check you are on the right dose. There are some foods and drinks that you should avoid, and certain diet changes may affect the dose you need. Your doctor or nurse will discuss these with you before starting on warfarin.

There isn't space in this Manual to list all the hundreds of medicines that are available, and new ones are being developed all the time.

If you have been given pills to take that are not listed on these pages, ask your doctor, pharmacist or facilitator what they are for.



Here is a summary of the main medicines and what they can be used for. It may be helpful to fill in the table with the names of the medicines you take, and how often you take them. Anything you are unsure about you can ask your doctor or facilitator.

	Reduce risk	Treat risk factors	Treat angina	Improve heart as a pump	Treat the heart rhythm	I take this type of medicine (tick)	Drug name	How many times a day?	Questions/ Comment
Antiplatelet agents	✓								
Beta-blockers	✓		✓	✓	✓				
Statins	✓	Cholesterol							
ACE inhibitors or ARB's	✓	Blood pressure		✓					
Nitrates			✓						
Calcium-channel blockers		Blood pressure	✓		✓				
Diuretics		Blood pressure		✓					
Anti-arrhythmics					✓				
Anticoagulants	✓								
Other									
Example: Antiplatelet						✓	Aspirin 75mg (low dose)	Once in the morning	Take it with or after breakfast (not on an empty stomach)with a glass of water



Medicines (continued)

- Prescription medicines come with an information sheet which contains important information about dosage, possible side-effects, what to do if you miss a dose, etc. Always be sure to read the information sheet right through when you start taking the pills.
- If you are not sure how to get your medicines when you need a new supply ask your GP, they may be able to set you up with a repeat prescription. This means you don't need to see the doctor every time you need more of your prescribed medicines.
- Make sure you always have enough medication and order your new prescriptions in time, so you won't run out before the new supply arrives. Sometimes this can take a few weeks, so try to plan ahead.
- Try to get into a routine of what you take and when. This means you will be less likely to miss a dose.
- If you find it hard to open your medicine bottles or get the tablets out of the blister-packs, let your facilitator know. They can discuss what options are available to help with this.
- If you have any questions or concerns about side effects, or taking any of your tablets, talk to your pharmacist, doctor or facilitator, there may be other options or medicines that will help with the side-effects.

You can write down any other questions or concerns you may have about your medicines in the space below:

Name of medicine: _____

Questions: _____

Name of medicine: _____

Questions: _____

Name of medicine: _____

Questions: _____



Chest pain

Chest pains are quite common in the early weeks after a heart attack.

They don't usually mean that you are having another heart attack. Other things can cause chest pain, too.

Think back to before your heart attack.

If you are like most people, you may have felt unusual pains in your body – from time to time. People who haven't had a recent heart attack usually put this sort of thing down to 'twinges', muscle cramps or creaking joints.

The pains are often caused by pain messages which travel up special nerves to the brain. Sometimes these nerves produce 'false alarms' which may be our body's way of testing the system.

It's not surprising if your body decides to test the system more often after a heart attack. And it's not surprising if you notice pains and twinges more than you did before. Don't worry about these pains – they will settle down in time.

This section of the Manual tells you more about some of the other common causes of chest pain after a heart attack. Some chest pains have physical causes, while others are caused or made worse by worry and anxiety.

Anxiety can make these pains worse in all kinds of ways:

- It causes muscle tension, which can lead to pain.
- It makes you much more aware of twinges and false alarms.
- It causes the adrenaline level in your blood to increase, and this can cause chest pain on its own.
- It can make you breathe too fast, causing chest pain (see 'Hyperventilation' section on page 148).

The wrong kind of breathing can cause chest pain if you breathe with your chest muscles. Look back at page 33, and remember that there are special breathing exercises on the relaxation CD/audio to help with this.

So with all these possible causes of chest pain, it's easy to see how you can get into a vicious circle.

You feel a pain in your chest – you worry about it – the worry makes it feel worse – you worry more – you get dizzy and sweaty – and so on. It's not surprising that some people end up back in hospital!

If you are getting pains in your chest, try to remember all of these things. And make sure that you are doing your relaxation and breathing properly, as this will help.

However, if the pain is severe, or just like the pain you experienced when you had your heart attack, follow the advice on page 35.



Angina

What causes angina?

Angina is a pain in the chest that occurs when the coronary arteries can't supply enough blood to part of the heart muscle to allow it to work properly.

It's a type of **cramp in the heart muscle. Just like a cramp in any muscle, it passes with time and doesn't do any lasting damage.**

Some patients experience angina as a tightness or discomfort in the chest rather than a pain. Occasionally it maybe felt in the arm, the back or the jaw. While others may experience shortness of breath.

In Parts 1 and 2 of the Manual we explained how **coronary artery disease** causes your arteries to become narrowed by the build-up of cholesterol and scar tissue on the artery walls, and how one of these diseased areas became inflamed and blocked by platelets, causing your heart attack.

Coronary artery disease can also cause angina.

Your heart is just like any other muscle. The harder it works, the more oxygen it needs. Oxygen is carried in the bloodstream to the heart via your coronary arteries.

If your arteries are healthy, they can provide your heart with as much oxygen as it needs. They can stretch to let more blood and therefore oxygen through when your heart is working harder than normal. Diseased arteries can't do this.

Remember what we said on page 46 about rubbish being washed down a river. The rubbish builds up at certain points. Where the river gets narrower, the water flows slower through the narrow parts, but it still gets through just the same. But as the river gets narrower still, it may reach a stage where it begins to dry up downstream.

It's the same inside your arteries. You already know that when a narrow region of an artery becomes completely blocked, the blood can't get through at all, and the result is a heart attack.

However, before an artery becomes entirely blocked, so much rubbish may become stuck at a narrow part that your narrowed arteries might not be able to supply all of the blood and oxygen that your heart needs.

You may not notice this at first. But there may be times when your heart is working harder than usual (for instance, when you are running for a bus or walking uphill) and it lets you know about the problem. **That's angina.**



Angina (continued)

What brings on angina?

Most of the time angina is brought on by exercise and eased by rest. If your chest pain comes on when you exercise and goes away when you stop, it could be angina.

In some cases, angina can come on when you are not exercising. Strong emotions can bring it on, too.

As we have said before, emotions cause the release of adrenaline into the blood. This makes the heart beat faster. The faster it beats, the harder it works. The harder it works, the more oxygen it needs, so the more angina you get. The more angina you get, the more anxious you become. The more anxious you become, the more adrenaline you produce ... get the idea?

Imagine that you go out for a walk and your angina comes on sooner or worse than you expected. It feels a bit like the pain you experienced when you had your heart attack. So you think that you might be having another heart attack and you get worried. This causes you to tense up, and the adrenaline makes your heart beat faster.

It's easy to see how a vicious circle can be set up.

What can I do about angina?

In the last section we said that angina can be brought on by exercise and strong emotions. So you might think that in order to get rid of it you should stop taking exercise and avoid getting upset.

Wrong! That would be easier said than done – and anyway it doesn't work. If you stop exercising, you become less fit and it takes more effort to do even simple things. Avoiding exercise is the surest way to make angina worse.

Avoiding stress is virtually impossible, too – and remember that some stress and excitement is a normal part of life.

So, to keep angina in check:

- Keep up your exercise programme to gradually build up fitness. If you get angina when you are exercising, slow down or stop. As you get fitter, this should happen less often.
- Remind yourself that angina is not like a heart attack. It is simply your heart's way of saying 'Hold on a minute – I need to catch up.'
- If the pain comes on when you are upset, check that you are breathing correctly, using your stomach muscles and not your chest. Do your relaxation and check that you aren't tense. Wait for the pain to ease off, and congratulate yourself on dealing with it calmly.

- You may have been given a GTN spray to use. This can help to stop angina more quickly. Carry the spray with you at all times, and use it as well as doing the things listed above. If you are about to do something which you know will bring on angina, use your spray before starting.

Avoiding exercise and stress doesn't work. However, managing exercise and stress can be part of your treatment, and you can take some precautions. For instance, **your heart has to work harder on cold, windy days**. So if you are out and about, wrap up warm and try to keep sheltered from the wind. Some people have found that even breathing in cold air can bring on pain. If you find this is so, try wearing a scarf over your mouth.



Angina (continued)

Angina can also be worse **after a heavy meal**, when your heart has to pump a lot of extra blood to your stomach to help your digestion. This means two things. First, your heart has to work hard for an hour or so after the meal. Secondly, there's less blood for your muscles or your brain (that's why you feel sleepy!). So try not to exercise after a meal, when your heart is busy and your muscles are short of blood. In general, smaller meals are better.

Some people experience pain **at a particular point in a walk**. If this happens, try to break the cycle by varying your walk or by distracting yourself. Look around and work out how many things you can see that you never noticed before. Enjoy the walk and don't look for your angina.

Try this plan:

- Decide how far you can walk before getting angina (for example, to the corner shop).

- When you reach that point, slow down or stop and do your quick relaxation to get as calm as possible.
- Once you are relaxed, decide where your next stop will be and then carry on walking.

If you follow this simple plan you will find that the distance between stops gradually gets longer.

Will it go away?

In many cases angina will ease on its own. If it comes on after a heart attack, this may simply be because the heart attack has upset the balance of the blood supply to your heart muscle.

In a few weeks or months, as your heart recovers and you follow the advice in the Manual, the angina may well disappear completely. Remember what

you learned about the heart's amazing powers of recovery.

Finally, although we have concentrated on what **you** can do for angina, there is also a great deal that **doctors** can do.

This ranges from prescribing medicines (see 'Medicines' section on pages 135–142) to performing an operation to

widen the arteries (see page 150), or a bypass operation (see page 151).

It is important that you don't ignore angina. Tell your doctor or nurse about it – particularly if it is getting worse rather than better, or if it suddenly changes.



Breathlessness

Normally we don't think about our breathing. We just do it. But if we run for a bus, we breathe faster and we notice it – we feel breathless.

This can be because you are actually breathing faster, or simply because you are noticing your breathing more.

Some heart attacks affect the way that your heart works as a pump, and this may make you breathless. Usually this is

treated with medicines soon after the heart attack. If you are taking tablets for this, discuss it with your facilitator or doctor.

If the problem continues, your cardiologist may try further tests to find out whether your breathlessness is the kind that needs treatment.

But by far the commonest reason for feeling breathless is simply the fact that **you are noticing your breathing more.** Increased

awareness of breathing should settle within a few weeks after your heart attack but, just like chest pain, it is made worse by anxiety. So work on your relaxation and breathing.

One particular type of breathlessness requires a little more explanation. It is called **hyperventilation.**



Hyperventilation

On the relaxation CD/audio we told you about the best way to breathe. Hyperventilation is one type of wrong breathing, and it can cause chest pain. Often people who are rushed into hospital with a suspected heart attack turn out to be hyperventilating or having a panic attack (see Margaret Anderson's story on page 157).

Some people hyperventilate all the time and some only do so when they are under stress.

Very often the person who is hyperventilating is not aware of the problem, but those close to them may notice that they often sigh, or that their breathing often seems very fast, or that they sometimes stop breathing and then take a deep breath.

Another sign is that their chest is often heaving up and down. Because hyperventilation is difficult for you to spot in yourself, ask someone you

know well if they notice you doing any of these things.

Breathing like this makes you feel nervous, frustrated or twitchy. This is because raised adrenaline levels which are common after a heart attack can trigger hyperventilation.

Here are other symptoms:

- feelings of light-headedness
- the sensation that things around you are far away or unreal
- a feeling of tingling or numbness in the mouth, fingers or tongue
- a feeling that you can't get enough air
- wobbly legs
- a feeling that you must 'get away' from something.

What can I do about it?

- Practise the breathing exercises on the CD/audio at least once a day.

- Try to check your breathing often during the day, and correct it if you are breathing with your chest.
- Get your partner to remind you to breathe with your stomach.
- When you are making an effort – such as opening a door, getting out of a chair, opening a tight jar, etc. – **try not to hold your breath**, as this often sets it off.

If you keep practising it will get better – but it may take quite a long time, because you've probably been hyperventilating for years. If, after several months of practising, the problem is no better, or if you are having real difficulties getting your breathing right, ask your GP or facilitator to refer you to a clinical psychologist.

Palpitation

This is another symptom which may simply be due to being more aware of your body. Palpitation means being aware of your heartbeat. Just as with breathlessness, you usually only notice it if your heartbeat is faster than normal.

You may notice it more for some weeks after your heart attack. Some medicines may affect your pulse and make you more aware of it, too.

Palpitation can be made worse by anxiety, because adrenaline increases the heart rate.

And of course your heart rate will increase when you exercise. If you are unfit, it may remain higher for a time even after you stop.

Sometimes palpitation can be due to a problem. You might be aware of your heart missing a beat or putting in an extra one.

This is not usually serious, but let your doctor or facilitator know if it seems to be happening a lot.

If your heart has periods when it starts racing or the beat doesn't seem to be regular, let your doctor or facilitator know. You might need more tests or a special type of medicine called an anti-arrhythmic to control the problem.





Hospital tests

This section tells you about some of the routine tests you might expect when you are in hospital or when you go back for a follow-up appointment after your heart attack.

Exercise tolerance test (ETT or treadmill test)

The [exercise tolerance test \(ETT\)](#) is usually carried out using a motor-driven treadmill and a special ECG machine. You walk on the treadmill while your heart rate, blood pressure and ECG are being recorded. Every 3 minutes the treadmill speeds up or gets steeper so that you have to work harder and your heart rate and blood pressure increase.

The test is used to find out how well you are recovering physically and whether you have angina. The information that it provides can also help the rehabilitation team to advise you on further exercise.

Echocardiogram (Echo)

This is just like the ultrasound test that is performed on pregnant women. It is completely painless. Sound waves are used to produce a video of the heart beating. This allows the doctor to see how well your heart is pumping and to assess how the scar is healing.

Coronary angiogram (catheter study)

Depending on how your recovery is progressing and the results of the above two tests, the doctor may ask for an 'angio.' This test is used to find regions where the coronary arteries are diseased. A fine tube called a [catheter](#) is threaded up from the artery in your arm or leg into your heart. A special dye which shows up on X-rays is injected through the tube into the coronary arteries and filmed using an X-ray camera placed over your chest. A local anaesthetic is used to numb the area.

The X-ray pictures help your doctor to decide what, if anything, needs to be done to improve the blood supply to your heart muscle. The test takes about half an hour to complete, and you may have to lie flat in bed for a few hours.

Computerised tomography (CT) angiogram

This procedure uses an intravenous dye, usually iodine based, and CT imaging to scan the coronary arteries. Unlike the coronary angiogram, this is a non-invasive procedure. (no catheters are inserted into the arteries) Due to the accuracy of the images, it is a major new tool in the diagnosis of coronary artery disease, and can be helpful in cases where treadmill test results are unclear.



Treatments

This section tells you about two operations, either of which your cardiologist might recommend. The aim of each is to improve the blood supply to your heart muscle. These operations will only be performed if your angina cannot be controlled by taking medicines, or if your cardiologist considers that your arteries are too badly diseased to be left alone.

Angioplasty (Stent)

This is often called the 'balloon' treatment. From the patient's point of view, it is rather like the angiogram – in fact it is often done at the same time. This treatment may have been carried out when you were first admitted to hospital following your heart attack (see page 13).

The cardiologist will ask for your permission to go ahead and 'stretch' any narrowings that they find during the angiogram. A special type of catheter that has a small balloon on the end is threaded through the artery in your arm or the top of your leg and up into your coronary arteries. Using the X-ray camera, the balloon is positioned in the narrow part of the coronary artery. It is then inflated with enough pressure to stretch the artery where the coronary disease has narrowed it. The balloon is deflated and dye is injected to see whether the narrowing has been stretched wider. The balloon may need to be inflated several times in order to get the best result.

The cardiologist may decide to put a small metal tube called a [stent](#) inside the artery to make sure that it stays open after the angioplasty. The stent can be stretched by the balloon to fit the exact size and shape of the inside of the artery. Once it has been stretched it acts like scaffolding to hold the artery open while it repairs itself. Eventually the stent becomes embedded in the lining of the artery, helping to prevent it from becoming narrowed again in future.

Angioplasty usually requires at most an overnight stay in hospital.



Treatments (continued)

Coronary artery bypass graft (CABG)

This is a very common procedure, with thousands of operations being performed in the UK each year. The CABG is carried out by opening up the chest through the breastbone.

The idea is to use a different blood vessel to re-route the blood around the narrowed artery. There are two ways of doing this.

Running down either side of the breastbone are two arteries called the [internal mammary arteries \(IMAs\)](#), which normally help to supply blood to the chest wall. They run very close to the coronary arteries, and the surgeon can free the end of an IMA and stitch it on to the coronary artery downstream from the narrowing. You then have a new blood supply!

The second option involves removing a blood vessel from your leg or arm which is used to bypass the narrowed section of artery. The surgeon stitches one end to the main artery leading from your heart, just next to where your own coronary arteries start. The other end is then stitched to the coronary artery downstream from the narrowing.

A CABG operation requires about a week in hospital.

Heart surgery is advancing all the time, and your doctor may offer you another form of revascularisation (re-routing) which is not described in this manual. We would therefore encourage you to seek out more information from your cardiologist.



Anxiety

It's normal to worry after you have had a heart attack. Who wouldn't worry in the same situation? Obtaining more information about worry or anxiety can help many people to handle it better. That is what this section is about.

Anxiety can cause some or all of these physical feelings:

- dry mouth
- 'butterflies' or tightness in the stomach
- a cold sweat
- rapid pulse
- palpitations
- rapid breathing
- cold, clammy hands and feet
- tense, sore muscles
- a sensation of being far away or distant from what's going on
- feeling faint
- strange pains
- tingling in the feet and hands
- ... and many other feelings.

Anxiety has these cognitive or 'mental' effects:

- difficulty in concentrating
- difficulty with remembering
- difficulty with decision making
- racing thoughts
- problems with sleeping
- lack of self-confidence
- feeling that your personality has changed
- feeling that you are going mad.

Anxiety can cause these kinds of behaviour:

- not listening to what people are saying
- feeling unsatisfied
- restlessness
- fidgeting
- bad temper
- flaring up at little things
- losing your sense of humour
- becoming overcritical or sensitive
- becoming socially isolated.

We all get like this when we are worried, scared or under stress. The primitive emotion responsible is fear.

What's happening in my body?

When we are in danger or under stress, our bodies automatically produce chemicals to help us to cope.

The main chemical that we produce is **adrenaline**. In a real emergency it can help to save our lives.

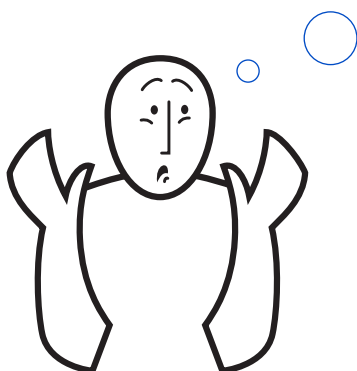
Our bodies make adrenaline to help us in an emergency.

Imagine that you are crossing the road. Suddenly a lorry comes hurtling towards you and this thought flashes through your mind:

This lorry isn't going to stop. I'll be killed!

The thought would probably be so quick that you wouldn't notice it. It triggers the production of adrenaline, and your adrenaline level goes up. The adrenaline gets you ready in an instant to take avoiding action.

It isn't only frightening events that make us produce adrenaline. Frightening words, thoughts or memories trigger it, too. This is the same for everyone.



Anxiety (continued)



Adrenaline gets us ready to fight or run when we are in danger. This is what it does:

Effects on the mind

■ It makes us concentrate on the danger and ignore other less important things.

■ It makes our thoughts race – so that we can look for quick ways to escape.

■ It makes us ‘think the worst.’

All animals have this ‘fight or flight’ mechanism. It’s what helps them to survive.

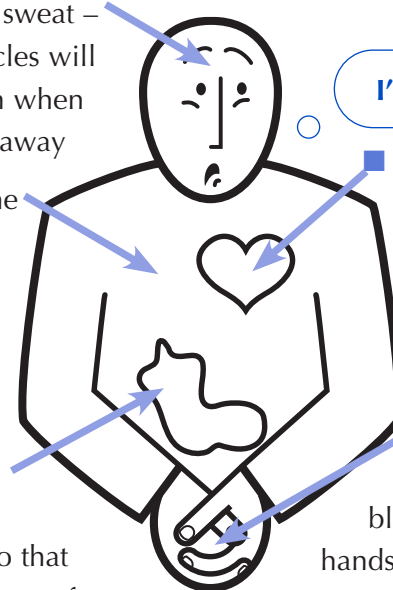
Another name for it is **fear**. If we didn’t feel fear we would be a danger to ourselves and to others.

Effects on the body

■ It produces a sweat – so that the muscles will be cooled down when we are running away

■ It increases the breathing rate, sending more oxygen to our muscles so that we can run faster

■ It shuts down the stomach – so that there is more energy for running



I’m going to die

■ It speeds up our heart rate – sending more blood to our muscles so that we can run faster

■ It reduces the blood supply to the hands and feet – so that there is more for the main muscles

The problems start when we make the adrenaline – but there’s no one to fight and we can’t run away (‘fight or flight’ syndrome)

A frightening thought, image or memory can produce adrenaline in the same way as a frightening event.

But this time there’s no one to fight and nowhere to run.

So what happens to the extra adrenaline? It doesn’t get used up, and it causes unpleasant physical and mental symptoms.

If we were running we wouldn’t notice these symptoms – but as we are probably just going about our normal routine, we do notice them.

The racing heart, cold sweat, dizziness and other symptoms make us feel scared and ill.

Dizziness
Racing thoughts

Cold sweat
Blurred vision

Rapid breathing
Muscle tension

Pins and needles
Cold, clammy hands

I’ve got a bad heart –
I might die

Dry mouth

Heart racing

‘Butterflies’ in the stomach



Anxiety (continued)

The anxiety story so far

Any worrying thought, memory or event triggers the production of adrenaline.

There is nothing wrong with this. It's part of nature's way of helping us to cope with danger so that we can live to fight another day.

If we are in real danger we do run or fight. We are too busy to notice the effects that adrenaline has on our body.

But what if a worrying thought happens when you are sitting in a chair, or trying to get to sleep, or shopping? It can cause very unpleasant feelings.

- It makes your thoughts race, so that you can't relax, or it may make it difficult for you to get to sleep.
- It increases your breathing rate – making you feel dizzy or faint and far away.
- It speeds up your heart rate – you may notice that your heart is racing or pounding.
- It shuts down your stomach, giving you a feeling of 'butterflies' or a knotted stomach or a sinking feeling.
- It produces an unpleasant cold sweat – it may make your forehead feel clammy.
- It shuts down the blood supply to your hands and feet – you may notice that they feel cold and clammy.

Why is this so common after a heart attack?

Do you remember the Stress Thermometer? It shows how you feel at different levels of adrenaline.

The more you relax the lower it gets.

Panic ✗
Very scared ✗
Scared ✗
Worried ✗
Bad-tempered ✗
Tense ✗
Excited ✗
Alert ✓
Wide awake ✓
Calm ✓
Sleepy ✓
Very relaxed ✓



These sensations will not harm you. They are just unpleasant. They will go away as you recover from your heart attack and realise that you are getting better.

As you know, having a heart attack can be very frightening.

This means that your adrenaline level is already raised.

Any worrying thought or event can be enough to put your Stress Thermometer level up from alert to worried – or higher.

However, relaxation lowers the level.

The vicious circle

If you notice these bodily feelings, this can lead to thoughts like:

- 'There is something wrong with my heart again.'
- 'I'm not getting better – I'm getting worse.'
- 'I feel faint – I may be going to pass out.'
- 'If I don't get to sleep soon I'll be ill.'
- 'I'm having another heart attack.'

Of course these thoughts will produce more adrenaline, so the bodily feelings get worse.

If we notice this, we worry even more – this makes the feelings worse, and so on.

This won't hurt you, but it's very unpleasant, especially if you don't know why it's happening! Occasionally, in some people, this vicious circle can go on quickly getting worse and worse until the adrenaline builds up to a very high level and they have what is called a **panic attack**.





Anxiety (continued)

In a full panic attack most people are convinced that they are dying.

However, a panic attack is not dangerous and it is usually over within 10–30 minutes.

This is because after a while the body runs out of adrenaline.

The person who is having the panic attack begins to notice that they are getting better.

They think:

'It's stopping – I'm safe.'

'It wasn't another heart attack after all.'

'I'm going to be all right.'

'I'm feeling better all the time.'

Thoughts like these switch off the production of adrenaline.

They slow down the pulse, calm the breathing, reduce the heart rate and stop the stomach churning – and the person feels better and better.

Many people experience sudden increases in the amount of adrenaline in their blood.

It isn't just something that happens to people who have had heart attacks.

It causes a feeling of being unwell, or feeling scared or angry.

This can be because a picture, a memory or even a smell has brought a worrying thought into the person's mind.

After a heart attack your adrenaline level is already quite high. These thoughts produce even more adrenaline.

The extra adrenaline raises the temperature on your Stress Thermometer, making you feel worried, tense or bad-tempered.

The feelings may quickly pass, or you may end up feeling tense and unhappy most of the time.

The medical name for this is **anxiety**.

It is not dangerous – just unpleasant.

It is common in people under any kind of stress – such as divorce, serious money problems, unemployment, etc.

It is very common in people who have had a serious illness.

We know from research that these thoughts almost always stop after a while.

Until they do, you may find that thoughts about your heart and illness are always in your mind.

'I feel exactly like that, but I've never noticed myself having sudden worrying thoughts.'

Very often these thoughts can pass through your mind so quickly that you don't notice them. What you do notice are the effects that the thoughts set off.

You can tell that this has happened if you suddenly feel:

- worried
- tense
- scared
- bad-tempered
- 'butterflies' in your stomach
- cold sweats
- weak
- tired
- sad
- breathless
- any of the other symptoms of anxiety.

These very quick thoughts, of which we are hardly aware, are called **negative automatic thoughts**.

Learning the true facts about heart disease and heart attacks can help to fight these thoughts.

On the next three pages are some examples of how anxiety has affected other people after a heart attack.



Anxiety (continued)

Deepak Singh's story

Anxiety can cause bad temper.

After his heart attack, Deepak Singh had returned to his work as storeman at a large garage.

His job involved fetching parts from the storage racks and then filling out paperwork. Sometimes he had very little to do. At other times he would have a rush of work, with people queuing up waiting for him to get the parts they needed.

He found that he was getting very bad-tempered with the other people at work.

Three or four times he had suddenly felt very ill and had been sent home.

After a while he began to hate his work, and he dreaded going in each day.

When he wasn't at work he felt fine, except when he thought of having to go to work.

Soon his dislike of work meant that he was taking a lot of time off.

He said that he hated all the people there and he wanted to give up work.

He felt that he wasn't strong enough for it any more.

He went to see his doctor who, after doing a number of tests, told him that there was no longer anything physically wrong with him.

When the effects of anxiety were explained to Deepak, he realised that he became bad-tempered mainly when there was a long queue of people all waiting for him to serve them. Using a simple method that he had been shown, he was able to discover the automatic thoughts that occurred each time the queue built up.

These thoughts were:

'Having all these people waiting is putting me under stress.'

'Stress is bad for the heart.'

'It could cause another heart attack, which could kill me.'

It is not surprising that Deepak was getting annoyed.

Although he hadn't been aware of it, he was really thinking:

'These people could kill me!'

Knowing what his automatic thoughts were didn't cure Deepak. He still found himself getting angry and anxious whenever a queue built up.

But he did get better, and this is how:

■ First he learned relaxation, and he practised relaxing every evening.

■ This brought down the temperature on his Stress Thermometer. It made him feel a bit calmer during the day.

■ While he was at work, if he noticed any symptoms, such as 'butterflies' in his stomach or a cold sweat, he used the 'rapid relaxation' technique.

■ This relaxed his muscles, which helped to reduce the tension.

■ To stop the negative thoughts (about stress damaging his heart), he repeated to himself in his mind:

'I'm quite relaxed, which is good for my heart.'

'The more I relax, the better I feel.'

After a few weeks of this he felt OK again, and he was soon working happily with no more worries about stress.



Anxiety (continued)

Margaret Anderson's story

If you get tired or breathless or feel your heart beating fast, it can make you think that there is something wrong with your heart. This can cause anxiety, which makes you feel much worse.

A good example of this is what happened to Margaret Anderson. She had a close and loving family.

When she had a heart attack, they were all very shocked and scared, especially her husband, John.

John took over a lot of the things that Margaret had once done, including the shopping.

He often told her that shopping at the supermarket would be too much for her.

Margaret was actually very fit (probably fitter than John, who took very little exercise).

One day, she decided that she would do the week's shopping to show John that she could manage again.

She went to the local supermarket, which was very crowded.

She began to feel rather short of air and could feel a slight sweat on her forehead.

After she had filled her trolley, she found that the queue at the checkout was very long.

As she stood in the queue, she began to feel as if things around her were miles away, like looking down the wrong end of a telescope.

This feeling frightened her. She also noticed that her heartbeat was very fast.

She thought that this might mean she was having another heart attack.

By now she was desperate to get out of the shop, but she didn't want to draw attention to herself by just walking away and leaving her trolley.

She hardly heard a thing that the woman at the checkout said to her, and she can't remember how she got out of the shop.

All she could think was:

'I'm having another heart attack.'

Somehow she managed to phone her husband at work.

He drove her to the hospital. After a brief chat about the symptoms Margaret had felt, the doctor was able to tell her that she had had a panic attack.

She told Margaret about the physical symptoms of anxiety.

Margaret realised that she had been feeling rather worried before she went into the supermarket.

Most people feel a little anxious in crowded places, and this had also raised her adrenaline level.

Once she thought about it, she remembered that shopping in that particular store had often made her feel uncomfortably hot and short of air because it was overheated.

Because she had once had a heart attack, she had mistaken these feelings.

This had automatically made her think:

'Maybe John is right – I'm not strong enough to do this.'

'Maybe it will be bad for my heart.'

'Maybe I'll have another heart attack and die.'

This had produced even more adrenaline – enough to make her feel sick.

Then she had noticed her heart racing. This made her think:

'There is something wrong with my heart. I can feel it!'

With these thoughts going round and round in her head, she produced more and more adrenaline, which eventually caused a panic attack.

The same kind of thing often happens if a person feels tired suddenly, or notices a strange pain or new sensation.



Anxiety (continued)

Edward Maxwell's story

A common experience is to suddenly feel anxious or tired or unwilling when you think about having to do something.

This is how anxiety affected Edward Maxwell.

Before his heart attack, Edward had been a keen gardener.

While he was in hospital and during the first few weeks after he returned home, nothing got done in the garden and it began to look a mess.

After a couple of months Edward was driving again and fit enough to walk 5 miles every day.

But he still did nothing about the garden.

Although he often intended to start, whenever he looked at the messy state of the garden he felt weak and tired, and put it off.

Eventually even hearing the garden mentioned, or catching a glimpse of it out of the window, made his stomach churn and his heart speed up.

After reading about anxiety, he realised that every time he thought about the garden, these negative automatic thoughts were going through his mind:

'I ought to weed the garden.'

'Weeding might be too much for my heart.'

'It might cause another heart attack and I might die.'

In other words, any thought of the garden made him think of having another heart attack and dying.

Edward's wife often remarked that he seemed to have lost interest and that he should start gardening again.

These negative thoughts were triggering off the adrenaline that was making him feel unwell and tired – and making him avoid the gardening.

Once he realised what the thoughts were, he was able to notice this happening and he corrected the negative thoughts by thinking calming thoughts instead. He thought:

'I ought to weed the garden.'

'So long as I take it carefully and stop if I get too tired, it won't hurt me.'

'In fact it may make me fitter, which would be good.'

Thinking about it in this new way reduced the anxiety, and he was soon gardening again. After a while he was enjoying it more than ever.

Anxiety: summing up

It is normal to feel anxious after a heart attack or any other very frightening experience.

This anxiety usually gets less as the memories of the heart attack fade.

It is caused by a raised adrenaline level. This is normal after a heart attack.

It can affect us in two ways:

1 The raised adrenaline level may cause a general feeling of tension that is present all the time.

It may make sleeping difficult.

We may feel generally restless and unable to concentrate on anything. We may find that our anger flares up quickly, or we may seem cut off and become distant.

It may make us unwilling to go out, take exercise, etc.

2 We may feel all right most of the time, but get sudden feelings of 'butterflies' in the stomach, weakness, or a feeling of dread as if something terrible is about to happen.

This is usually because of an automatic negative thought or memory, generally about the danger of having another heart attack. Our body can't tell the difference between a real danger (such as a lorry heading straight towards us) and a fear about having a heart attack. It just does its job and produces more adrenaline. This can result in our suddenly feeling ill, peculiar or scared.

Although this is unpleasant, it is not dangerous.

Of course we may experience both 1 and 2.





Anxiety (continued)

Some solutions

Sudden anxiety

As soon as you notice yourself getting anxious, think about the statements below. They are called the Six Rules for Coping with Anxiety.

You may like to cut out the box below and carry it with you.

Cut along dotted line 

Cut along dotted line 

Six Rules for Coping with Anxiety

- 1 It is not my heart or mind that is going wrong.
- 2 The feelings that I have are the effects of adrenaline.
- 3 These feelings are not dangerous – just unpleasant.
- 4 Notice what is happening in my body right now – not what I fear may happen.
- 5 Remember that when I stop adding to my anxiety with negative thoughts, it starts to fade by itself.
- 6 Wait, and give the fear time to pass. Do not fight it or run away from it. Just accept it.

Doing any of the rapid relaxation techniques that work for you will help the feelings to go more quickly.



Anxiety (continued)

See if you can identify the negative thought that has produced the sudden rise in adrenaline.

These thoughts happen very quickly.

The best thing to do is to try and remember what you were thinking about just before the feelings appeared.

For example, in Edward Maxwell's case the thought was 'I ought to weed the garden.'

At first he didn't realise why this thought should be a problem, so he used the 'So Why' test.

The 'So Why' test.

'I should weed the garden.'



'So why' should that worry me?

'It might be too much for my heart.'



'So why' should that worry me?

'I might have another heart attack.'



'So why' should that worry me?

'I might die.'

This is also called the downward-arrow technique, because each 'So why' gets you further down towards the negative automatic thought.

Here is another example from Deepak Singh.

'There is a queue building up.'



'So why' should that worry me?

'It puts me under pressure.'



'So why' should that worry me?

'Pressure is bad for the heart.'



'So why' should that worry me?

'I might have another heart attack.'



'So why' should that worry me?

'I might die.'

Cut along dotted line

Use the 'So Why' test

Step 1

'What was I thinking about, looking at or remembering just before I started feeling like this?'

Step 2

 'So why' should that worry me?

Answer:

 'So why' should that worry me?

Answer:

 'So why' should that worry me?

Answer:

Cut along dotted line

